

Building RAG document Q&A with nvidia NIM and Langchain

Difference between Langchain and Nvidia NIM

These two sit at completely different layers, so comparing them directly can feel confusing at first.

One-line difference

- **NVIDIA NIM** → runs AI models as **ready-to-use APIs (inference layer)**
 - **LangChain** → helps you **build AI workflows/apps (orchestration layer)**
-

Side-by-side comparison

Feature	NVIDIA NIM	LangChain
Type	Inference service (microservices)	Application framework
Main role	Serve models via APIs	Build pipelines (RAG, agents)
Runs models?	✓ Yes	✗ No
Provides workflows?	✗	✓
Focus	Performance + deployment	App logic + integration
Typical user	Infra / ML engineers	App developers

What NVIDIA NIM actually is

NIM (NVIDIA Inference Microservices) gives you:

- Pre-packaged AI models (LLMs, vision, etc.)
- Optimized to run on NVIDIA GPUs
- Exposed as **API endpoints**

👉 Think:

“Run high-performance models easily (locally or in cloud)”

You call it like an API:

Your app → NIM endpoint → model output

What LangChain actually is

LangChain helps you:

- Connect LLMs + databases + tools
- Build:
 - RAG pipelines
 - Chatbots with memory
 - Agents

👉 Think:

“How do I structure my AI application?”

How they work together

This is the key relationship:

LangChain (logic)



NVIDIA NIM (model API)



Response

Example idea:

- LangChain → handles retrieval + prompts
 - NIM → runs the LLM
-



Simple analogy

- **NVIDIA NIM** = engine 🚗
- **LangChain** = navigation system 🗺️

You need both:

- Engine → to move
 - Navigation → to know where/how to go
-



Important distinction

- NIM competes with:
 - OpenAI API
 - Groq
 - Hugging Face inference endpoints
 - LangChain works **on top of all of them**
-



When to use what

Use NVIDIA NIM if:

- You want **fast, optimized model serving**
 - You have NVIDIA GPU infrastructure
 - You need production-grade inference
-

Use LangChain if:

- You're building:
 - RAG apps
 - Chatbots
 - AI workflows
-



Bottom line

- NVIDIA NIM = “run models efficiently”
- LangChain = “build apps using those models”

If you want, I can show a **real code example using LangChain + a NIM endpoint** so you see how they plug together.

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nvidia > Main Menu Shop Drivers Support 🔍

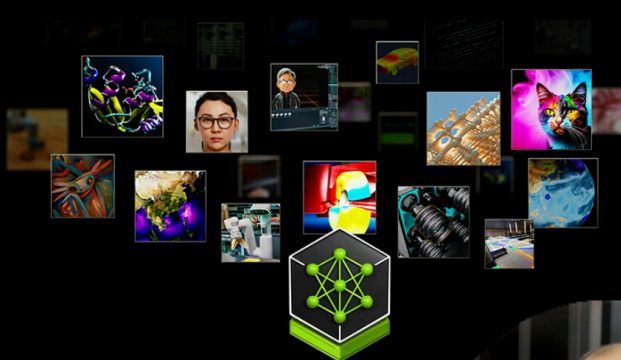
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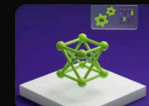
The top large language models for your enterprise AI



meta / **llama3-70b-instruct** **PREVIEW**

Powers complex conversations with superior contextual understanding, reasoning and text generation.

chat language generation



mistralai / **mistral-8x7b-instruct-v0.1**

An MOE LLM that follows instructions, completes requests, and generates creative text.

advanced reasoning chat



mistralai / **mistral-large** **PREVIEW**

Excels in complex multilingual reasoning tasks, including text understanding, and code generation.

chat code generation



google / **gemma-7b** **PREVIEW**

Cutting-edge text generation model text understanding, transformation, and generation.

chat language generation

Fresh Off the Press

The latest innovations in intelligence models



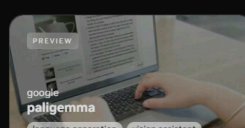
ibm / **granite-34b-code-instruct** **PREVIEW**

chat code generation



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google / **paligemma** **PREVIEW**

language generation vision assistant



seallm / **seallm-7b-v1** **PREVIEW**

chat large language model

The top screenshot shows the NVIDIA Llama3-70b-instruct model page. The page features a header with the NVIDIA logo, a search bar, and navigation links for 'Explore' and 'Docs'. A user profile dropdown menu is visible, showing the name 'Krish Naik', a 'Sign Out' button, and '954 Credits left'. The main content area includes a 'Back' link, the model name 'meta / llama3-70b-instruct' with a 'PREVIEW' badge, and a description: 'Powers complex conversations with superior contextual understanding, reasoning and text generation.' Below this are tabs for 'chat', 'language generation', 'large language models', and 'text-to-text'. A 'Model Card' tab is selected, displaying a warning about AI model outputs and a 'Preview' button. A chat interface shows a conversation where the user asks 'how are you' and the model responds. To the right, a 'Python' code editor shows a script using the OpenAI API to interact with the model. The bottom screenshot shows a code editor with two files: 'requirements.txt' and 'app.py'. The 'requirements.txt' file lists the following dependencies: openai, python-dotenv, langchain_nvidia_ai_endpoints, langchain_community, faiss-cpu, streamlit, and pypd.

https://build.nvidia.com/meta/llama3-70b

Search NVIDIA AI

Explore Docs

Back

meta / llama3-70b-instruct **PREVIEW**

Powers complex conversations with superior contextual understanding, reasoning and text generation.

chat language generation large language models text-to-text

Model Card

API Reference

AI models generate responses and outputs based on complex algorithms and machine learning techniques, and those responses or outputs may be inaccurate or indecent. By testing this model, you assume the risk of any harm caused by any response or output of the model. Please do not upload any confidential information or personal data. Your use is logged for security.

Preview JSON

Reset Chat

Hi! It's nice to meet you. Is there something I can help you with, or would you like to chat?

how are you

I'm doing well, thanks for asking! I'm a large language model, so I don't have feelings or emotions like humans do, but I'm always happy to chat with you and help with any questions or topics you'd like to discuss. How about you? How's your day going so far?

Python Node Shell

```
from openai import OpenAI

client = OpenAI(
    base_url = "https://integrate.api.nvidia.com/v1",
    api_key = "$API_KEY_REQUIRED_IF_EXECUTING_OUTSIDE"
)

completion = client.chat.completions.create(
    model="meta/llama3-70b-instruct",
    messages=[{"role": "user", "content": "how are you"}],
    temperature=0.5,
    top_p=1,
    max_tokens=1024,
```

Welcome requirements.txt app.py

requirements.txt

```
1 openai
2 python-dotenv
3 langchain_nvidia_ai_endpoints
4 langchain_community
5 faiss-cpu
6 streamlit
7 pypd
```

Rest of the code is self explanatory